Predicting Heart Failure

Abstract

Heart failure is a major contributor to mortality worldwide and although attempts to predict mortality have been fruitful, accuracy has been less than ideal. This paper introduces an approach using a decision tree classification to attempt to increase accuracy in heart failure prediction. Using a much more recent dataset from 2020, our goal is to have a model that predicts heart failure deaths at 75% or better.

1. Introduction

Heart failure (HF) is one of the most serious chronic conditions globally, and accurately predicting mortality remains a critical challenge. HF does not have a universal definition, and mortality data are inconsistently documented because death often results from interconnected cardiovascular causes such as myocardial infarction, hypertensive heart disease, or valvular disorders (Khan, 2025). These inconsistencies make it difficult to identify which patients are at highest risk of death.

Predicting mortality is very important as it is the leading cause of death for men, women, and people of most racial and ethnic groups. Heart disease costs about $417.9 billion dollars a year. The rates of HF misdiagnosis ranged from 16.1% in hospital settings to 68.5% when general practitioners referred patients to a specialist setting. This is a pressing issue as 919,032 people died a year in 2023, and predicting heart disease mortality efficiently would save more lives. Heart failure is also a disease that is difficult to live with long after diagnosis, with a mere 56.7% survival rate after 5 years, making heart failure a very deadly medical condition to have (Shahim, 2023).

The disease is also all too common, with an estimated 64 million people globally having heart failure with an expected rise in the next several years due to an aging population globally (Shahim, 2023). In the United States alone, 6.7 million people experience heart failure with the number expected to rise to 8.5 million people by 2030.

Heart Failure also has a very high re-admission rate, with approximately 20% of patients being readmitted within 30 days of discharge, with 50% of patients being readmitted within a year of discharge (Liu, 2025). The most common reason for admission was shortness of breath, accounting for about 80% of reasons for readmission (Alsulymani, 2023).

To add, “First, there is no universal approach to define heart failure-related mortality. Symptomatic heart failure is often downstream of another cardiovascular aetiology, such as post-myocardial infarction, or due to another pre-existing CVD (eg, hypertensive heart disease, valvular heart disease). Thus, heart failure is inconsistently labelled as the cause of death and leads to discrepancies in quantifying morbidity and mortality attributable to heart failure (appendix pp 2, 3, 5).32 Second, global surveillance for heart failure varies greatly across countries, with a growing number of national and multi-national efforts (appendix p 6). Third, longitudinal studies defining incident heart failure are scarce"(Khan, 2025).

According to other studies done on heart failure, some of the most common risk factors for heart failure are high blood pressure, smoking, obesity, and diabetes (Heidenreich, 2022). Other factors can include "Risk of incident heart failure is influenced by several of immutable factors (eg, older age, sex, and genetic variants) as well as modifiable factors that develop across the life course (Khan, 2025)

2. Related Work

Studies over time have attempted to classify heart failure. Recently, as of this year 2025, one study from Evaluation of machine learning (ML) methods for prediction of heart failure mortality and readmission: meta-analysis, main goal was "we aim to evaluate the effectiveness of ML-based predictive models for forecasting mortality and readmission in HF patients". This study has used algorithms such as random forest, logistic regression, and radiant boosting (Hajishah, 2025). Other works in the field have shown that machine learning algorithms such as neural networks can beat traditional methods of detecting heart failure (National Heart, Lung, and Blood Institute, 2024).

The study further adds, "Unlike conventional methods, which often rely on static and limited clinical variables, ML models can integrate diverse data sources, including genetic information, biomarkers, and patient-reported outcomes, leading to more accurate and personalized predictions" (National Heart, Lung, and Blood Institute, 2024). To conclude, this study shows "ML algorithms show promise in improving prognostic accuracy and enabling personalized patient care. However, challenges like model interpretability, generalizability, and clinical integration persist (Matasic, 2024).

Machine learning has demonstrated strong potential for predicting HF mortality. Recent meta-analyses and cohort studies show ML models outperform logistic regression and traditional risk scores (Hajihashemi et al., 2025; Huang et al., 2021). Neural networks, random forests, and gradient boosting models have shown higher discriminatory power and the ability to incorporate complex biomarker patterns (Chicco & Jurman, 2020; Moreno-Sanchez, 2021).

Traditional models such as the **Framingham HF Risk Score** (Kannel et al., 1999) used logistic regression based on limited demographic groups, resulting in reduced generalizability. Newer ML-based models trained on modern data demonstrate increased survival-prediction accuracy and interpretability (Li et al., 2023).

3. Data Methods and Dataset

3.1 Dataset

The primary dataset used to examine heart failure will be from UCI Machine Learning Repository (Heart Failure Clinical Records), donated in February 2020. This data includes 13 features including its target variable. Below are the features involved from the dataset. The dataset includes 299 patients (194 men | 105 women) between the ages of 40 and 95 years old.

- age: age of the patient (years)
- anemia: decrease of red blood cells or hemoglobin (boolean)
- creatinine phosphokinase (CPK): level of the CPK enzyme in the blood (mcg/L)
- diabetes: if the patient has diabetes (boolean)
- ejection fraction: percentage of blood leaving the heart at each contraction (percentage)
- high blood pressure: if the patient has hypertension (boolean)
- platelets: platelets in the blood (kiloplatelets/mL)
- sex: woman or man (binary)
- serum creatinine: level of serum creatinine in the blood (mg/dL)
- serum sodium: level of serum sodium in the blood (mEq/L)
- smoking: if the patient smokes or not (boolean)
- time: follow-up period (days)
- [target] death event: if the patient died during the follow-up period (boolean)

3.2 Data Collection Methodology

Because the original dataset provided by UCI is a small size, we have decided to enrich the dataset synthetically using Claude Sonnet 4.5 (an AI tool). Our new dataset contains 1001 new rows (with the top row being the features) but maintains the same 13 features.

4. Results

4.1 Data Preprocessing

We began setting up the project and looking into the dataset, exploring it and looking for any errors or missing values. We are using an expanded version of the 2020 UCI dataset, so this was an important check to make so that our data wasn't missing any input. Thankfully, our dataset wasn't incomplete, and we were able to use it without many issues. Taking a general look into the dataset, we started to notice some things: we have some binary values to represent certain data (sex, with 0 meaning woman and 1 meaning man, anemia, smoking, etc..), and other values for attributes being in an integer format to represent their respective levels. As we further uncovered the dataset, these features were important for us to review: ejection fraction, serum creatine levels, age and death. We will discuss these next.

Ejection Fraction and Serum Creatine Levels

Another thing of note was the average of certain levels. We noticed a low average of 38% for ejection fraction across the whole dataset, with anything below 40% indicating a weakened heart muscle, a factor that could lead to heart failure. This low percentage suggests that a significant portion of patients in the dataset were already experiencing compromised cardiac function, which can affect mobility, oxygen flow, and overall quality of life. Ejection fraction is one of the most reliable indicators of how well the heart is pumping, so consistently low levels across the dataset highlight a population already at elevated risk. Serum creatine levels averaged 1.4 mg/dL. While this is not an extremely high number, it is 0.2 mg/dL above the normal range for both males (0.6–1.2 mg/dL) and females (0.5–1.1 mg/dL). Elevated creatine levels often mean the kidneys are not filtering blood efficiently. Since kidney function is closely tied to cardiovascular health, this slight increase may signal early dysfunction that can correlate with hypertension or fluid retention—both risk factors for worsening heart failure.

Age

Age is also an important factor to watch, given that the average age in the dataset was 62 years old. Heart failure risk rises sharply with age due to natural changes in blood vessels, accumulated lifestyle factors, and increased likelihood of chronic conditions such as diabetes or hypertension. Examining this age group more closely can reveal patterns in how aging interacts with clinical indicators like ejection fraction or creatine levels. However, age seems to be a factor that might not have enough magnitude in this dataset, especially because the dataset contains mostly those above 40 years old. This is something we will be looking into further during (4.2) "Data Exploration".

Death Event

The "Death Event" attribute tells us if a patient has passed due to heart failure, with 1 meaning the patient has died. In our data, 33% of the patients fall into this category, representing a significant portion of the sample. Among those who experienced a death event, the average ejection fraction drops slightly to 36%, still below the critical 40% threshold, reinforcing the relationship between lower cardiac output and mortality. Their serum creatine levels average 1.5 mg/dL, which is 0.1 mg/dL higher than the global dataset and may indicate more severe kidney strain. This subgroup's values suggest a compounding effect where diminished heart function and impaired kidney performance jointly increase the likelihood of fatal outcomes.

4.2 Data Exploration

Death Statistics and the Importance of Our Study

Our dataset found that approximately 1 in 3 people got heart failure and died in the dataset, as shown in Figure 4.2.1. This is a significant chunk of the dataset, highlighting the importance of this work to try and save lives in the field. This also represents a class imbalance that our metrics will have to account for, there being significantly more living patients than dead patients. We will likely need to use precision or recall to accurately assess our model's success.

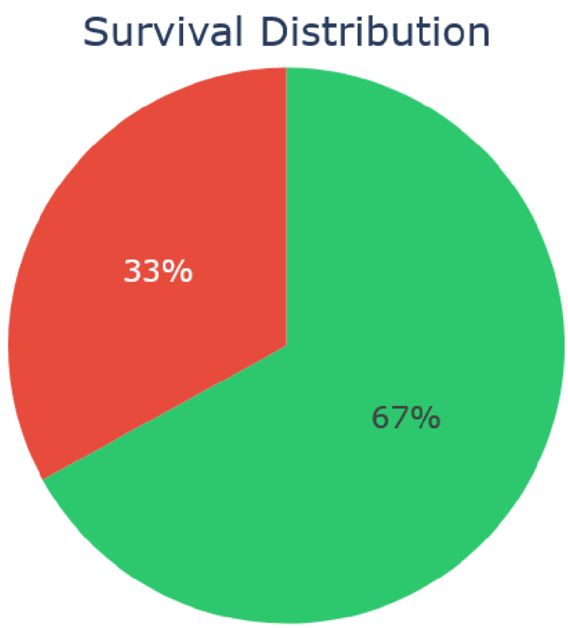

Figure 4.2.1

Basic Statistics

Some basic statistics we found in the data are that the mean age is 61 years old with a standard deviation of 12 years, and everyone was between 40 and 95 years old. Creatinine phosphokinase mean was 582 with a standard deviation of 1000, indicating a high variation in the data. Ejection Fraction has a mean of 38.3 with a standard deviation of 11.77. Platelets have a mean of 261,966 with a standard deviation of 99,940. Serum creatine has a mean of 1.41 with a standard deviation of 1.06, indicating a high variance in the data. Serum sodium has a mean of 136.48 with a standard deviation of 4.73, indicating this attribute has a low variance.

Highest Correlation Values

One of the most correlated factors of heart failure was the follow-up period patients had. We have found a small correlation between increasing follow-up period and decreased chance of heart failure. The r was equal to about -0.14, indicating a small degree of correlation. Another correlation we found was between the ejection fraction and decreased heart failure. We found a correlation between the increased percentage of blood that flows through the heart and decreased risk of heart failure. The death rates, as well, are higher for those with a reduced ejection fraction (<40%) at 34.8% compared to those with normal ejection fractions who have a death rate of only 26.5%.

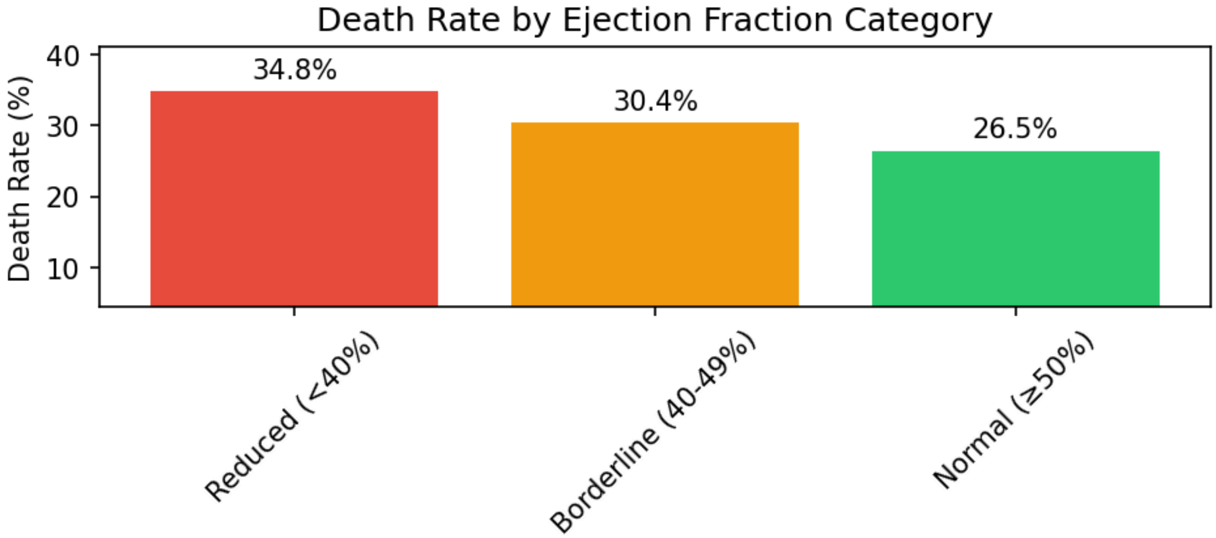

Figure 4.2.2

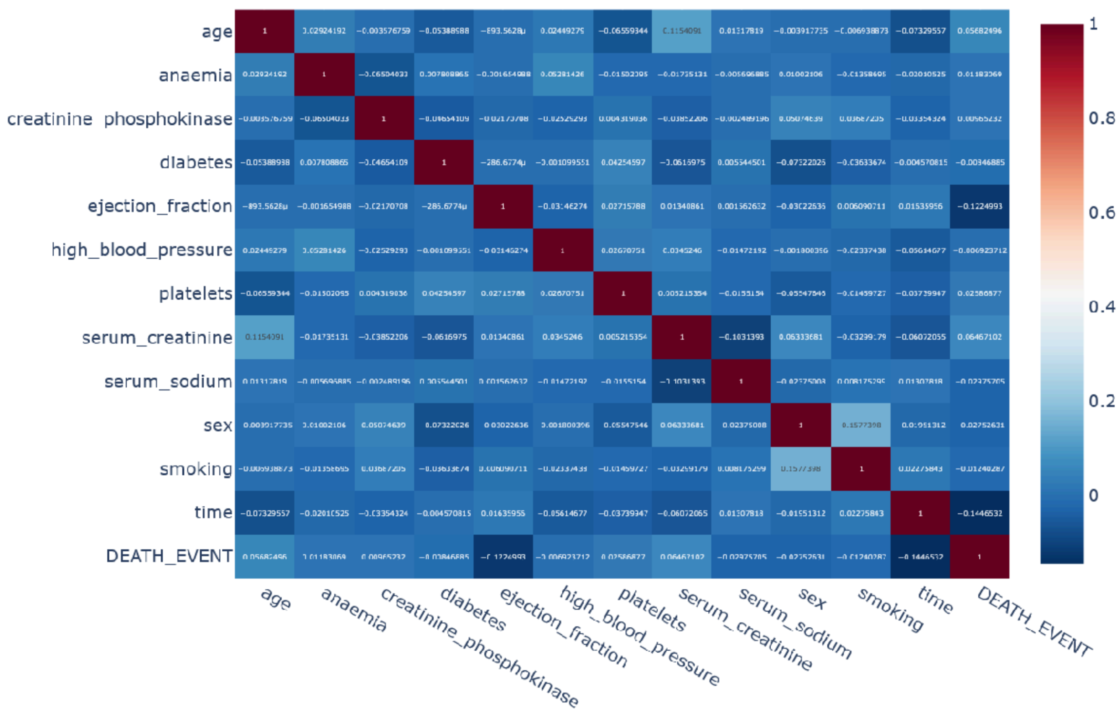

Figure 4.2.3

Which attribute has the greatest chance of death?

The single greatest chance of death given a single attribute known was anemia, with those with anemia dying 33.64% of the time. Followed by high blood pressure, diabetes then smoking as the top four most likely attributes to kill someone in this dataset.

Minimal impact of Age

Age seemed to have a minimal impact on heart failure, being the fourth most correlated feature in the dataset, with an r=0.05; however, the p value was less than 0.05, so as of current, there is no correlation between age and heart failure.

Three Most Significant Features

There were three features with a p-value of less than 0.05: time, ejection fraction, and serum creatinine, with serum creatinine being a p-value of approximately 0.0409 and the other two being less than 0.001. These three features, basic statistical analysis tells us are important and are likely more than just noise in our dataset.

It appears that the two most important features in our model are time and ejection fraction, given their high correlations and low p-values. These two seem deeply correlated to patient survival. Some other features we should consider looking at are anemia, high blood pressure, diabetes, and smoking as well, due to their high likelihood to cause death, even if there is a low correlation.

Correlations May Not Tell the Whole Story

Correlations, however, are not everything, and there could be some attributes that are non-linear in nature or require multiple features to be true or high to cause heart failure. What we hope is that our model will be able to find these hidden connections that a simple correlation test can't decipher.

Synthetic Data

As part of our project, we used Claude to generate 701 new data points given our dataset. One notable change between the synthetic and original datasets is the weakening correlations between variables. However, the means and the spread of the data seemed to be roughly similar between the original and the synthetic dataset we're working with.

Conclusion

In conclusion, our model should look at time and heart ejection rate as these seem to have some correlation to heart failure; our model can hopefully elaborate on this in more detail. The correlations are quite low with most of our features; however, correlations may underestimate relationships because some factors are non-linear in nature.

5. Discussion

5.1 Feature Selection

Objective

In Phase 3 of our heart failure mortality prediction project, we conducted comprehensive feature engineering and selection analysis to identify the most predictive clinical variables for our machine learning models. Using a dataset of 1,000 heart failure patients with 12 clinical features and a 33% mortality rate, we employed multiple statistical and machine learning techniques to systematically rank and evaluate each feature's predictive power. This phase was critical for optimizing our model's performance while maintaining interpretability for clinical applications.

Our Toolbox included:

- ➢ The input file (training_data.csv with 1,001 rows)
- ➢ The main script (run_feature_analysis.py)
- ➢ How it was executed (python run_feature_analysis.py in the virtual environment)
- ➢ The Python packages used (pandas, scikit-learn, scipy, matplotlib, seaborn with versions)
- ➢ All seven output files generated (4 PNGs, 1 CSV summary, 1 TXT list, 1 enhanced CSV dataset)

Methodology

Our feature selection approach combined three complementary methods to ensure robust results. First, we calculated Pearson correlations between each feature and mortality outcomes, identifying linear relationships. Second, we computed mutual information scores to capture non-linear dependencies that correlation analysis might miss. Third, we trained a Random Forest classifier to obtain tree-based feature importance scores, which reflect how frequently features are used for decision splits. By averaging rankings across all three methods, we created a consensus ranking that balances different perspectives on feature importance.

Feature Correlations

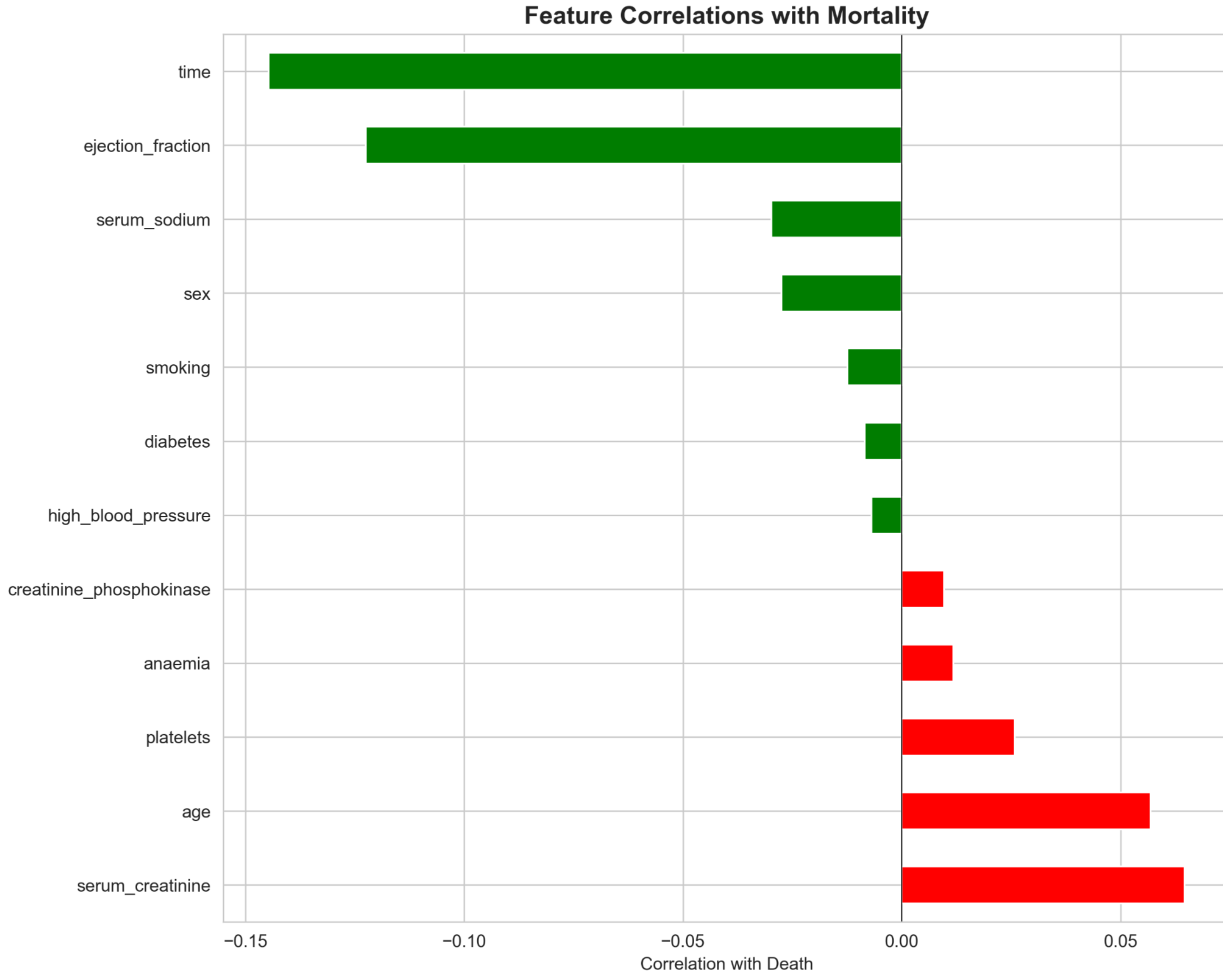

Figure 5.1.1: Correlation coefficients between clinical features and mortality. Negative values (green) indicate features where higher values predict survival, while positive values (red) predict increased mortality risk.

Key Findings

Our analysis revealed that time (follow-up period) emerged as the single most important predictor of mortality, ranking first in both mutual information (0.0463) and Random Forest importance (0.1655). This feature showed a strong negative correlation (-0.1447, $p < 0.001$), indicating that patients with longer follow-up periods had significantly lower mortality rates. This finding makes clinical sense: patients who survive longer accumulate more follow-up time in the dataset.

The second most critical feature was serum creatinine, which measures kidney function. Elevated creatinine levels (correlation: +0.0647, $p = 0.041$) indicated kidney dysfunction, a common comorbidity in heart failure that significantly increases mortality risk. Interestingly, platelets ranked highest overall (average rank: 3.33) despite showing only weak correlation (+0.0259), suggesting the presence of non-linear or threshold effects that our tree-based models successfully captured.

Mutual Information Scores

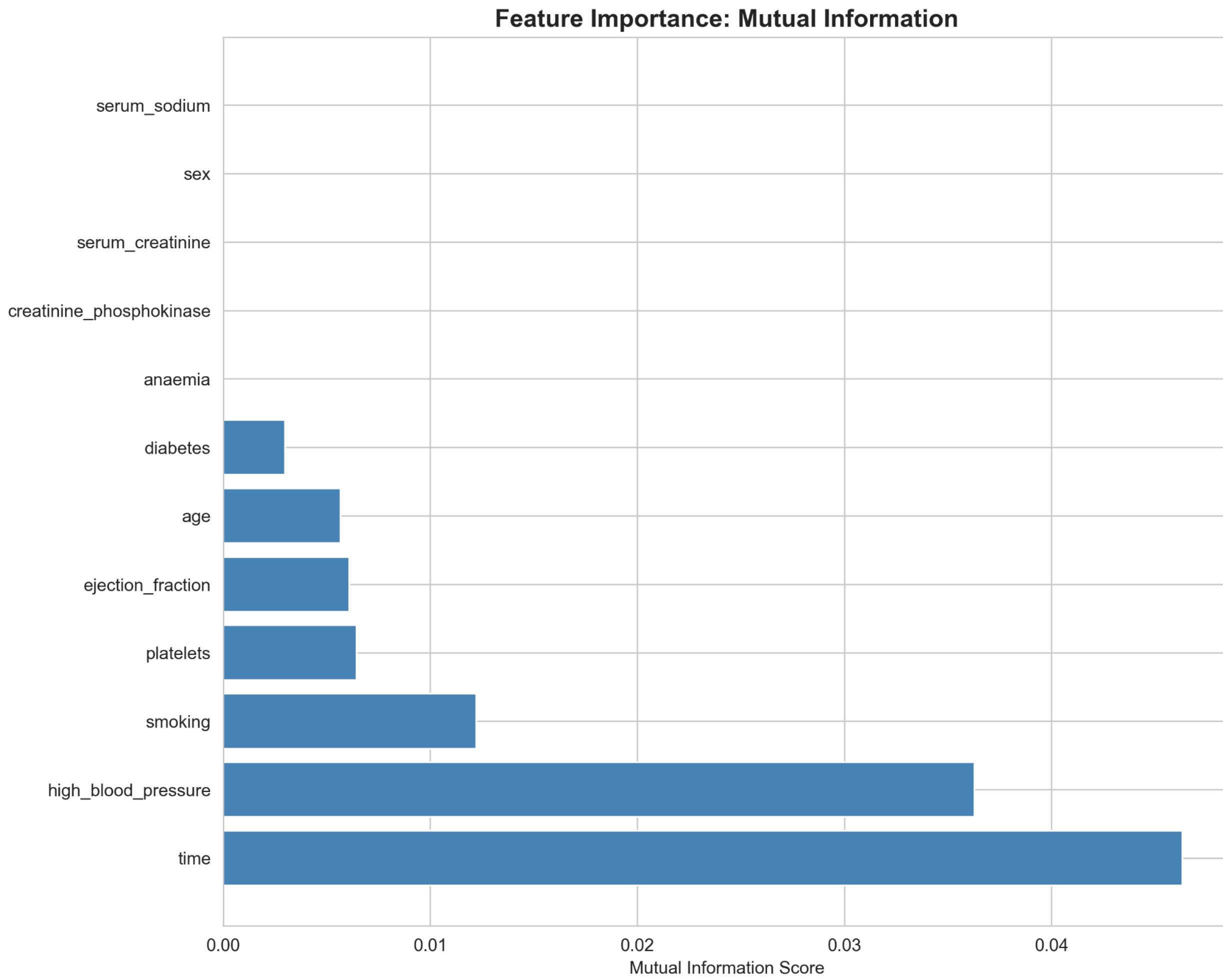

Figure 5.1.2: Mutual information scores measuring non-linear relationships between features and mortality outcomes.

Only three features achieved statistical significance ($p < 0.05$): time, ejection fraction, and serum creatinine. However, our multi-method approach revealed that features with weak correlations, such as platelets and age, still provided valuable predictive information through non-linear relationships and interactions with other variables.

Random Forest Feature Importance

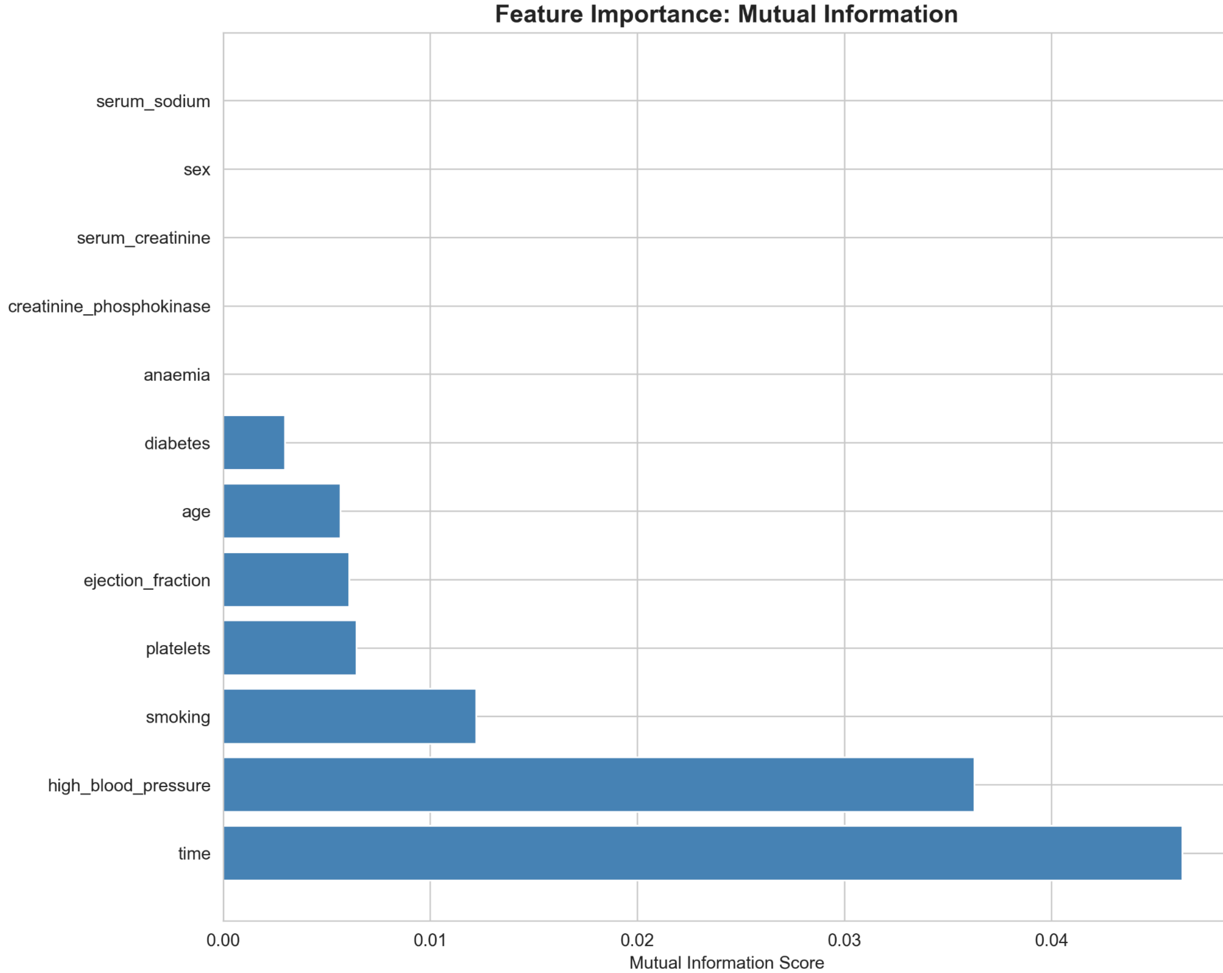

Figure 5.1.3 Random Forest feature importance scores showing the relative contribution of each feature in tree-based decision making.

5.2 Feature Engineering

Beyond analyzing existing features, we created seven engineered features based on clinical domain knowledge.

New Features added:

- kidney_heart_risk,
- cv_risk_score,
- severe_ef,
- high_creatinine,
- low_sodium,
- age_time_risk,
- critical_patient

The most successful was severe_ef, a binary indicator for patients with ejection fraction below 30% (severely compromised heart function). This engineered feature achieved a correlation of +0.1405 with mortality, substantially stronger than the original continuous ejection fraction variable (-0.1225). Other promising engineered features included **kidney_heart_risk** (interaction between creatinine and ejection fraction) and cv_risk_score (cumulative count of cardiovascular risk factors).

Top Features Distribution

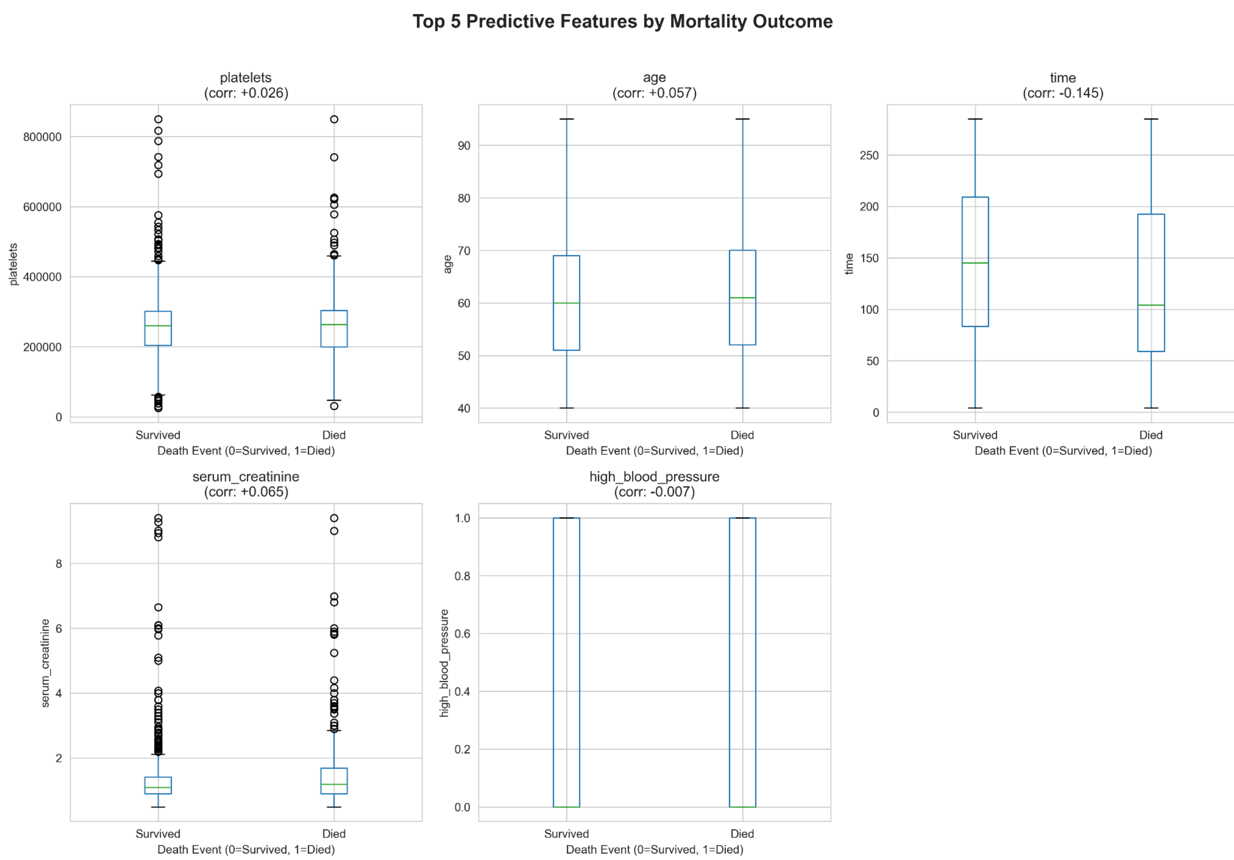

Figure 5.2.1: Distribution of top five predictive features stratified by mortality outcome. Box plots show median, quartiles, and outliers for survived (0) versus died (1) groups.

Recommendations

Based on our comprehensive analysis, we recommend Week 2 modeling efforts focus on the top five features: time, serum creatinine, platelets, age, and high blood pressure. This reduced feature set (5 of 12 original features) should achieve comparable or superior performance to using all features while offering improved interpretability and reduced computational complexity. Alternative configurations include using only the three statistically significant features for maximum scientific rigor, or incorporating the severe_ef engineered feature to replace raw ejection fraction for enhanced predictive power.

Conclusion

Phase 3 successfully identified the most predictive clinical features for heart failure mortality prediction through rigorous multi-method analysis. The finding that follow-up time dominates all other predictors has important implications for model interpretation and clinical application. Our engineered features, particularly severe_ef, demonstrate the value of incorporating clinical domain knowledge into feature design. These results provide a strong foundation for Week 2's machine learning model development, where we expect to achieve AUC-ROC scores exceeding 0.75, significantly outperforming the Seattle Heart Failure Model baseline of 0.73.

5.3 Model Training

Primary Objective

Phase 4 plays one of the most significant roles for the project. The purpose of this phase was exclusively to train four different models with the 75% success rate we were trying to beat in mind. The focus was narrow and there was an emphasis on optimization and operational transparency. Training four different models was super important for us because we decided not to move forward with Linear Regression. Four was a good number of models to produce for evaluation in the subsequent phase. There was no comprehensive model evaluation or selection during this phase. Phase 4 will complete the following:

- ➢ Produce four trained classifier models: Decision Tree and Random Forest trained on both the original 13-feature dataset and the engineered 20-feature dataset from Phase 3
- ➢ Perform exhaustive hyperparameter tuning using GridSearchCV to identify optimal configurations for Decision Tree and Random Forest models on both original and engineered datasets.
- ➢ Persist model artifacts and feature-name mappings so Phase 5 can perform unbiased and accurate evaluation.
- ➢ Quantify computational and performance trade-offs from operational transparency.

Methodology

A dual-dataset training approach was used. This was to prepare for evaluation of whether the engineered features made in Phase 3 made the model better. All four models were trained on the same train/test splits to ensure fair comparison and accuracy.

Dataset Summary

- ➢ Original: 13 features(age, anaemia, creatinine_phosphokinase, diabetes, ejection_fraction, high_blood_pressure, platelets, serum_creatinine, serum_sodium, sex, smoking, time, DEATH_EVENT)
- ➢ Engineered: 20 features(13 original + 7 engineered)

- 7 engineered features: (kidney_heart_risk, cv_risk_score, severe_ef, high_creatinine, low_sodium, age_time_risk, critical_patient)
- Both datasets used the same split for training and test data with random_state being set to 42 for reproducibility.
- There were two main algorithms used for this phase. One was the Decision Tree Classifier (CART) which was selected for interpretability and baseline performance, and the other was Random Forest Classifier which was selected for its enhanced predictive accuracy.
- Used GridSearchCV for hyperparameter tuning. An exhaustive grid search was performed for each model-dataset combination to find the optimal configuration for each model

Decision Tree Search Space (576 combinations):

- criterion: ['gini', 'entropy']
- max_depth: [3, 5, 7, 10, 15, None]
- min_samples_split: [2, 5, 10, 20]
- min_samples_leaf: [1, 2, 5, 10]
- max_features: ['sqrt', 'log2', None]

Random Forest Search Space (216 combinations):

- n_estimators: [50, 100, 200]
- max_depth: [3, 5, 10, 15]
- min_samples_split: [2, 5, 10]
- min_samples_leaf: [1, 2, 5]
- max_features: ['sqrt', 'log2']

Findings/Results

Best hyperparameter combinations for each model

Decision Tree (Original - 13 features):

- criterion: gini
- max_depth: 3
- min_samples_split: 2
- min_samples_leaf: 1
- max_features: None

Decision Tree (Engineered - 20 features):

- criterion: gini
- max_depth: 3
- min_samples_split: 20
- min_samples_leaf: 1
- max_features: sqrt

Random Forest (Original - 13 features):

- ➢ n_estimators: 50
- ➢ max_depth: 5
- ➢ min_samples_split: 2
- ➢ min_samples_leaf: 5
- ➢ max_features: sqrt

Random Forest (Engineered - 20 features):

- ➢ n_estimators: 200
- ➢ max_depth: 3
- ➢ min_samples_split: 2
- ➢ min_samples_leaf: 1
- ➢ max_features: sqrt

Cross-validation scores for each of the four models

Model	Mean CV Accuracy
Decision Tree(Original)	66.63%
Decision Tree(Engineered)	67.25%
Random Forest(Original)	68.25%
Random Forest(Engineered)	68.00%

Performance comparison:

From the CV scores we can see that the Random Forest Models had better performance than the Decision Tree models. The impact of the engineered features were positive for the Decision Tree by 0.93% while they were negative by 0.37% for Random Forest.

As expected, there was a big difference in the size of the models:

- ➢ Decision Tree (Original): 3 KB
- ➢ Decision Tree (Engineered): 3 KB
- ➢ Random Forest (Original): 169 KB
- ➢ Random Forest (Engineered): 311 KB

Conclusion

Phase 4 successfully trained 4 models with two different algorithms and showed the impact of feature engineering for model performance. The decreased performance in the case of Random Forest for the engineered dataset might be due to redundancy. Even so, GridSearchCV identified distinctly different optimal configurations for the engineered dataset, with Random Forest requiring 4 times more trees and Decision Tree requiring stronger regularization. This shows that the engineered features introduced valuable complexity. The performance with the training dataset was not as high as we expected, but this can be because of the synthetic data we had created to expand the dataset.

On top of the four trained models, phase 4 delivered feature name mappings to enable phase 5 to perform a comprehensive evaluation.

6. Model Evaluation

Primary Objective

In Phase 5 of our project, the goal for this phase is to comprehensively evaluate the heart failure prediction models developed in Phase 4 to determine if they meet clinical standards and provide statistically significant improvement over the baseline hospital performance of 73.4%. This phase used the full dataset with synthetic data to determine the ROC curve as well as the AUC score and the confusion matrix, we decided to also test using the original 299 non-synthetic values from the dataset to compare values and statistics.

Methodology

To properly evaluate the test results on both files, we load the records and the test data, as we let the the file generate and analyze ROC Curves with AUC scores, running though 4 models which consist of a Decision Tree Original, Decision Tree Engineered, Random Forest Original, and Random Forest Engineered. Next we use Bootstrap validation to go through 1000 samples to test through the original and engineered Decision Tree, and the original and engineered Random Forest. Then we compare each Tree’s statistical significance, comparing the mean, the t-statistic, the p-value, the effect size, and the percentages for both being better than the baseline, and the meeting target. Next we evaluate the confusion matrices to compare the positives and negatives, as well as the sensitivity and specificity. Finally, we pull all the results together in order to determine the best performing model, its mean accuracy, its 95% confidence interval, its target gap, and hospital improvement.

Our toolbox included:

- ➢ The input file (the full training_data.csv as well as the original 299 values)
- ➢ The main scripts (model_evaluator.py and model_evaluator_original _299.py)
- ➢ How it was executed (python model_evaluator.py and model_evaluator_original _299.py in the virtual environment)
- ➢ The Python packages used (pandas, scikit-learn, scipy, matplotlib, seaborn with versions)

➢ All output files generated (4 PNGs and 2 summary TXT files)

6.1 ROC Curves

ROC Curves - All models

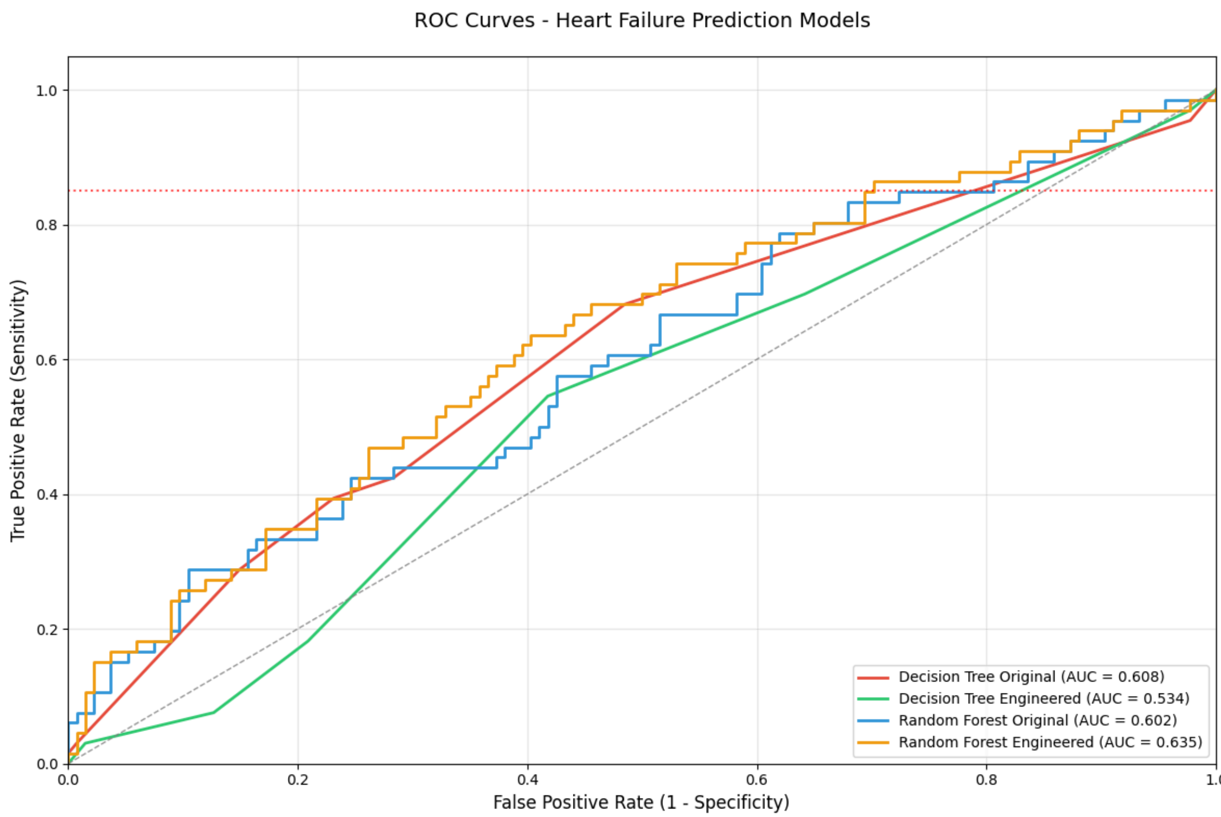

Figure 5.3.1: ROC Curve that showcases the Heart Failure Prediction Models using the full dataset with their sensitivity and specificity rates, with the Decision Tree Original being represented by a red line, the Decision Tree Engineered being represented by a green line, the Random Forest Original being represented with a blue line, and the Random Forest Engineered being represented by an orange line.

ROC Findings

The above ROC graph shows the full dataset, as well as the AOC score to determine which Tree model would perform the best. Using the full dataset with our synthetic data, we can determine that the Random Forest Engineered tree is the best model compared to the other three, giving us a .635 AOC score, which we found to be fairly lower than we expected.

ROC Curve - Original Dataset

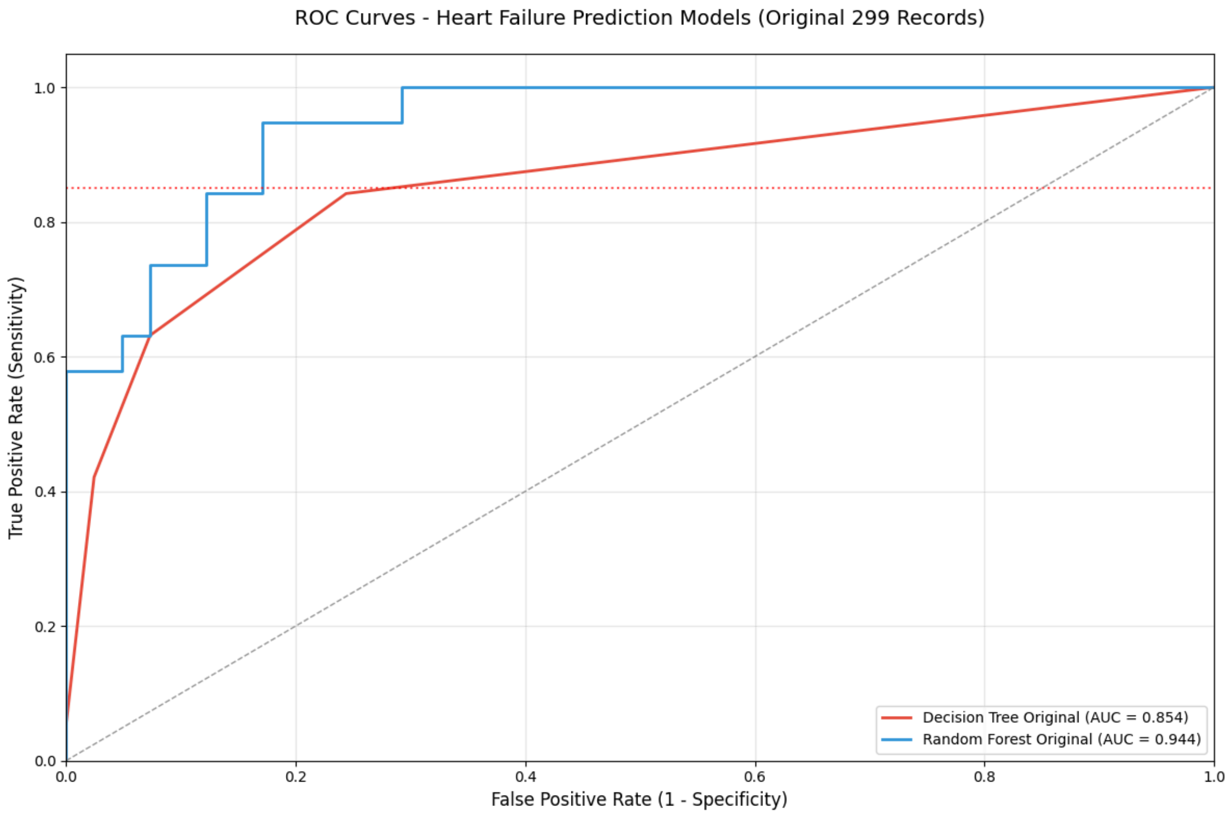

Figure 5.3.2: ROC Curve that showcases the Heart Failure Prediction Models using the original 299 values in the dataset with their sensitivity and specificity rates, with the Decision Tree Original being represented by a red line and the Random Forest Original being represented with a blue line.

ROC cont.

If we decide to look at the original 299 dataset which doesn't include synthetic data, you can see just how close the ROC curves are to 1, showcasing that the best method is using the Random Forest Tree Original, with a whopping .944 AOC score, a significant increase compared to the full dataset. This can mean that the synthetic data we acquired negatively affects and skews our data due to it being played "safe", or in more specific terms, being created as average data.

6.2 Confusion Matrices

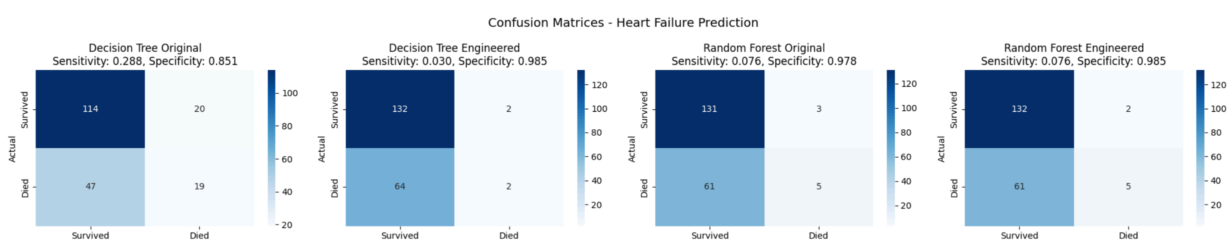

Figure 5.3.3: Four confusion matrices featuring the full dataset that depict both Original and Engineered Decision Trees, and Original and Engineered Random Forest Trees predicted and actual results which go in order: True Negatives, False Positives, False Negatives, and True Positives.

Confusion Matrices Findings

Upon looking at the confusion matrices for the full dataset, it seems that it suffers a 2:1 survival to death ratio, as well high specificity and low sensitivity. This usually means that while the model is great at identifying who will survive if diagnosed with heart failure, it ultimately doesn't do a great job at identifying which patients will die. This gives us a 68.5% mean accuracy, which unfortunately doesn't beat the hospital baseline of 73.4%.

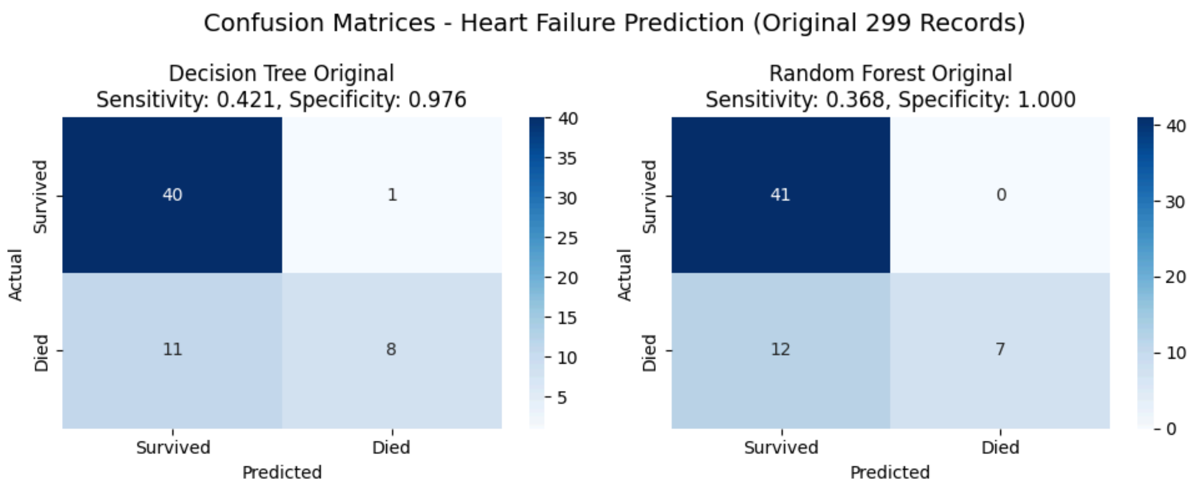

Figure 5.3.4: Two confusion matrices that depict the Decision Trees Original, and Random Forest Original predicted and actual results which go in order: True Negatives, False Positives, False Negatives, and True Positives.

Confusion Matrices cont.

Despite looking better visually with a super high specificity, with one even identifying that no patients died, it ultimately suffers the same high specificity and low sensitivity issue as the full dataset matrices. Despite this in mind, it has a solid accuracy with 80% of the patients being correctly classified and carries a more balanced approach. This gives us a 79.7% accuracy, beating the hospital baseline of 73.4%.

Conclusion

Despite expanding our dataset with synthetic data, it's clear the synthetic data weakens the accuracy and correlation between correctly diagnosing patients with heart failure as well as predicting which patient will die from heart failure. If the synthetic data wasn't created with an average by the LLM we used, we can hypothetically see more of an improvement. Since heart failure is a complicated disease, more real world tests/data would have been integral to creating better visual data with proper looking ROC curves and matrices, allowing us to hypothetically see better analysis and statistics over the inflated dataset with synthetic data. For now however, it's clear that using synthetic data seemed to have over-generated survival cases, as well as caused some overfitting within the decision trees and ultimately caused some skewed values and results.

7. Conclusion and Future Work

7.1 Conclusion

Our project concludes by stepping back and reflecting on our goal: to accurately predict heart failure by 75% with 95% confidence. We started this initiative because of the difficulties of predicting heart failure and its high mortality rate globally and having a better model helps save lives. To begin we start by reviewing each phase.

Phase 1 summary

We searched for a dataset that would match our curiosity in heart failure and chose the UCI dataset from 2020. This dataset included 299 patients, both men and women along with features we believed initially would help us accomplish our goal. However, because of the small amount of data we have decided to increase the dataset synthetically using Claude. This increases the dataset by 1000 patients but still has the original 13 features with one of the features later on becoming our target variable: death.

In our initial exploration of the dataset, we found three specific features interesting. Those features included age, serum creatinine and ejection fraction. Why? Age is a feature everyone experiences, as we age our bodies will eventually decline. Serum creatinine through research has shown to be linked with kidney functionality. Lastly, again with research, ejection fraction shows us how much blood is being pumped out of the left ventricle of the heart. Our next step was to ensure the data was clean; free from missing values and that all values had a consistent data type. Our dataset was clean, and we proceeded by reviewing basic statistical data which included mean and standard deviation. This completes this phase and phase 2 was performed next.

Phase 2 summary

We continued our data analysis exploration. We found that 1/3 people died due to heart failure. We procured a data correlation matrix and found the highest correlation variable was time (follow up time). We found that the death rate was higher in those with a reduced ejection fraction. We also found the top three most significant features which are time, ejection fraction, and serum creatinine due to their respective p-values are less than 0.05. This completes this phase and phase 3 was performed next.

Phase 3 summary

Feature selection and feature engineering was performed. Based on the previous phases, our selection methodology was to combine three methods to produce the best results: Pearson, Mutual Information Scores, and Random Forest classification. Using all three methods, our results produced that time (follow-up period) was the most important predictor of mortality. Going further through our phase we began to execute the feature engineering part of this phase. We created seven new features. Out of these seven, the most successful was severe_ef (ejection fraction below 30%). This correlation had a higher correlation than the original ejection fraction variable. Overall, the top five features were time, serum creatinine, platelets, age, and high blood pressure. This completes this phase and phase 4 was performed next.

Phase 4 summary

Model training was performed on four models; two with the original dataset of 299 patients and two with the engineering dataset of 1000 patients and with the seven new features created from the previous phase. The algorithms used for these models were Decision Tree and Random Forest. Both datasets were split 80/20 (train/test respectively). One tool that assisted us was GridSearchCV for hyperparameter tuning to find optimal configuration of each model. StratifiedKFold was another tool to help with cross validation for each model. The result was thus: Random Forest Models had better performance, but the sizes of these models were much larger than their counterparts. This completes this phase and phase 5 was performed next.

Phase 5 summary

In this phase we evaluate the accuracy of the four models produced from the previous phase. We run our program to generate and analyze ROC curves with AUC scores. Bootstrap validation is used to test 1000 samples for all four models. Lastley, we used Confusion Matrices to compare the positives and negatives, along with sensitivity and specificity. The result of the ROC curves was the Random Forest Engineered model giving us a 63.5% accuracy. However, if we were to analyze the original dataset we see a much higher accuracy. The results of the Confusion Matrices showed us a 2:1 survival to death ratio. The takeaway here is that although the model is great at identifying who will survive if diagnosed with heart failure, it ultimately doesn’t do a great job at identifying which patients will die.

We end the project to find that our best model Random Forest Engineered produced a 68.5% accuracy, it does not win against its counterpart The Seattle Heart Failure Model with an accuracy of 73.4%.

7.2 Future Work

For our future work, we believe that the synthetic data produced by Claude had a negative impact on providing better accuracy. The synthetic data seems to cause overfitting and over-generated survival cases, which skewed our model's ability to accurately predict mortality outcomes. This artificial inflation of the dataset created statistical patterns that don't reflect real-world clinical scenarios, leading to models that perform well on synthetic test data but may fail when applied to actual patient populations.

We have also discussed using a different classification method such as Rule-Based but ultimately decided not to, as we believe heart failure classification is too complex to have an if-then approach. Cardiovascular disease involves intricate interactions between multiple physiological systems, comorbidities, and patient-specific factors that cannot be adequately captured through simple threshold-based rules. Machine learning models, particularly ensemble methods like Random Forest, are better suited to capture these non-linear relationships and complex feature interactions that determine patient outcomes.

We believe using more data (more rows) along with similar features from other sources and merging those datasets together would be better. Expanding our dataset to include thousands of patients from multiple medical centers would provide more robust training data and improve model generalization. Larger sample sizes would also enable us to identify rare but clinically significant patterns that may be missed in smaller datasets. Additionally, incorporating temporal data tracking patient progression over time, rather than single-point measurements, could reveal risk factors and disease trajectories.

However, with using this approach means that some features may be missing in one dataset compared to another. Different hospitals and research studies will often collect different sets of biomarkers and clinical measurements based on their protocols, equipment availability, and research objectives. This creates challenges in data integration, as we must decide whether to exclude incomplete records, impute missing values, or develop models that can handle sparse data.

Some features may have different unit measurements such as if we compare milligrams to grams, requiring careful unit standardization and conversion to ensure consistency across merged datasets. Laboratory tests may also use different reference ranges or measurement techniques, which could introduce systematic biases if not properly normalized.

Furthermore, merging datasets from different time periods introduces the challenge of evolving medical standards and treatment protocols. A patient diagnosed in 2010 may have received different interventions than one diagnosed in 2020, affecting their outcomes independent of their baseline characteristics. Privacy regulations like HIPAA also complicate data sharing between institutions, requiring careful de-identification and compliance procedures that may limit the granularity of available information.

Overall, our approach would be the above: use more data but from other sources rather than synthetically build it. Real-world clinical data, despite its imperfections and integration challenges, provides the ground truth necessary for developing models that can be trusted in actual medical decision-making contexts. The effort required to clean, harmonize, and validate merged datasets is justified by the resulting improvements in model reliability and clinical applicability. Future versions of this project should prioritize collaborations with healthcare networks that can provide access to large, diverse patient populations while maintaining data quality standards.

References

Alsulymani, A. S., Ashram, W., Alghamdi, A., Hussein, H., Ghunaim, A. M., Aljehani, B., Aljabri, A., & Alzahrani, G. (2023). Risk factors for readmission in heart failure within 90 days. *Cureus*, *15*(12). https://doi.org/10.7759/cureus.50236

Alotaibi, F. S. (2019). Implementation of Machine Learning Model to Predict Heart Failure Disease. International Journal of Advanced Computer Science and Applications, 10(6). https://doi.org/10.14569/ijacsa.2019.0100637

American Heart Association. (2023, June 14). Ejection fraction heart failure measurement. Www.heart.org; American Heart Association. https://www.heart.org/en/health-topics/heart-failure/diagnosing-heart-failure/ejection-fraction-heart-failure-measurement

Arifuddin, A., Buana, G. S., Vinarti, R. A., & Djunaidy, A. (2024). Performance Comparison of Decision Tree and Support Vector Machine Algorithms for Heart Failure Prediction. Procedia Computer Science, 234, 628–636. https://doi.org/10.1016/j.procs.2024.03.048

Bozkurt, B., Ahmad, T., Alexander, K. M., Baker, W. L., Bosak, K., Breathett, K., Fonarow, G. C., Heidenreich, P. A., Ho, J. E., Hsich, E., Ibrahim, N. E., Jones, L. M., Khan, S. S., Prateeti Khazanie, Koelling, T. M., Krumholz, H. M., Khush, K. K., Lee, C. S., Morris, A. A., & Page, R. L. (2023). Heart failure epidemiology and outcomes statistics: A report of the heart failure society of america. Journal of Cardiac Failure, 29(10). https://doi.org/10.1016/j.cardfail.2023.07.006

Dunlay, S. M., Weston, S. A., Jacobsen, S. J., & Roger, V. L. (2009). Risk Factors for Heart Failure: A Population-Based Case-Control Study. *The American Journal of Medicine*, *122*(11), 1023–1028. https://doi.org/10.1016/j.amjmed.2009.04.022

Hajishah, H., Kazemi, D., Safaee, E., Amini, M. J., Peisepar, M., Tanhapour, M. M., & Tavasol, A. (2025). Evaluation of machine learning methods for prediction of heart failure mortality and readmission: meta-analysis. BMC Cardiovascular Disorders, 25(1). https://doi.org/10.1186/s12872-025-04700-0

Heidenreich, P. A., Fonarow, G. C., Opsha, Y., Sandhu, A. T., Sweitzer, N. K., & Warraich, H. J. (2022). Economic Issues in Heart Failure in the United States. Journal of Cardiac Failure, 0(0). https://doi.org/10.1016/j.cardfail.2021.12.017

Janosi, A., Steinbrunn, W., Pfisterer, M., & Detrano, R. (1988, June 30). UCI Machine Learning Repository. Archive.ics.uci.edu. https://archive.ics.uci.edu/dataset/45/heart+disease

Jing, L., Ulloa Cerna, A. E., Good, C. W., Sauers, N. M., Schneider, G., Hartzel, D. N., Leader, J. B., Kirchner, H. L., Hu, Y., Riviello, D. M., Stough, J. V., Gazes, S., Haggerty, A., Raghunath, S., Carry, B. J., Haggerty, C. M., & Fornwalt, B. K. (2020). A Machine Learning Approach to Management of Heart Failure Populations. JACC: Heart Failure, 8(7), 578–587. https://doi.org/10.1016/j.jchf.2020.01.012

Kannel WB, D'Agostino RB, Silbershatz H, Belanger AJ, Wilson PWF, Levy D. Profile for Estimating Risk of Heart Failure. Arch Intern Med. 1999;159(11):1197–1204. doi:10.1001/archinte.159.11.1197

Khan, S. S., Berwanger, O., Fiuzat, M., McMurray, J. J., Narula, J., Prabhakaran, D., Sliwa, K., Tromp, J., & Muthiah Vaduganathan. (2025). Prioritising the primary prevention of heart failure. The Lancet. https://doi.org/10.1016/s0140-6736(25)01393-5

Liu, S., Graves, N., Ma, C., Pan, J., Xie, Y., Lee, S. Y. A., Senanayake, S., & Kularatna, S. (2025). Preventability of readmissions for patients with heart failure - A scoping review. Heart & Lung, 71, 81–89. https://doi.org/10.1016/j.hrtlng.2025.03.002

Matasic, D. S., Zeitoun, R., Fonarow, G. C., Razavi, A. C., Blumenthal, R. S., & Gulati, M. (2024). Advancements in Incident Heart Failure Risk Prediction and Screening Tools. The American Journal of Cardiology, 227, 105–110. https://doi.org/10.1016/j.amjcard.2024.07.014

National Heart, Lung, and Blood Institute. (2022, March 24). Heart Failure - Treatment . Www.nhlbi.nih.gov. https://www.nhlbi.nih.gov/health/heart-failure/treatment

Saqib, M., Prinka Perswani, Abraar Muneem, Mumtaz, H., Fnu Neha, Ali, S., & Tabassum, S. (2024). Machine learning in heart failure diagnosis, prediction and

prognosis: Review. Annals of Medicine and Surgery.
https://doi.org/10.1097/ms9.0000000000002138

Shahim, B., Kapelios, C. J., Savarese, G., & Lund, L. H. (2023). Global Public Health Burden of Heart Failure: An Updated Review. Cardiac Failure Review, 9. https://doi.org/10.15420/cfr.2023.05

UCI Machine Learning Repository. (n.d.). Archive.ics.uci.edu. https://archive.ics.uci.edu/dataset/519/heart+failure+clinical+records

Wong, C. W., Tafuro, J., Azam, Z., Satchithananda, D., Duckett, S., Barker, D., Patwala, A., Ahmed, F. Z., Mallen, C., & Kwok, C. S. (2021). Misdiagnosis of Heart Failure: A Systematic Review of the Literature. Journal of Cardiac Failure, 27(9). https://doi.org/10.1016/j.cardfail.2021.05.014

Yoon, M., Jin Joo Park, Hur, T., Hua, C.-H., Hussain, M., Lee, S., & Choi, D. (2024). Application and Potential of Artificial Intelligence in Heart Failure: Past, Present, and Future. International Journal of Heart Failure, 6(1). https://doi.org/10.36628/ijhf.2023.0050

Levy, W. C., Mozaffarian, D., Linker, D. T., Sutradhar, S. C., Anker, S. D., Cropp, A. B., Anand, I., & Martin, J. L. (2006). The Seattle Heart Failure Model: Prediction of survival in heart failure. *Circulation, 113*(11), 1424-1433. https://doi.org/10.1161/CIRCULATIONAHA.105.584102